Assignment 3

Task 1 — Entity and Attribute Definition

HOSPITAL

Attributes: HospitalID (PK), HospitalName, StreetAddress, City, State, ZipCode, PhoneNumber

Composite attributes: Address = (StreetAddress, City, State, ZipCode)

Multivalued attributes: None

Entity type: Strong entity. HospitalID uniquely identifies each hospital.

DEPARTMENT

Attributes: DepartmentID (PK), DepartmentName, Location, PhoneExtension, HospitalID

Composite attributes: None

Multivalued attributes: None

Entity type: Strong entity. Department names are unique only within a hospital, but DepartmentID provides a globally unique identifier.

DOCTOR

Attributes: DoctorID (PK), FirstName, MiddleName, LastName, Specialty, Email, PhoneNumber, HireDate

Composite attributes: DoctorName = (FirstName, MiddleName, LastName)

Multivalued attributes: None

Entity type: Strong entity. DoctorID uniquely identifies each doctor.

PATIENT

Attributes: PatientID (PK), FirstName, MiddleName, LastName, DOB, Gender, StreetAddress, City, State, ZipCode, PhoneNumber

Composite attributes: PatientName = (FirstName, MiddleName, LastName); Address = (StreetAddress, City, State, ZipCode)

Multivalued attributes: None

Entity type: Strong entity. PatientID uniquely identifies each patient.

APPOINTMENT

Attributes: AppointmentID (PK), AppointmentDate, AppointmentTime, Reason, Status

Composite attributes: None

Multivalued attributes: None

Entity type: Strong entity. AppointmentID provides a globally unique identifier for each appointment.

MEDICATION

Attributes: MedicationID (PK), MedicationName, Manufacturer, UnitCost

Composite attributes: None

Multivalued attributes: None

Entity type: Strong entity. MedicationID uniquely identifies each medication.

Task 2 — Relationship Identification

Relationship	Entities	Cardinality	Participation	Notes
Hospital contains Department	Hospital–Department	1:N	Hospital = partial, Department = total	Each department belongs to exactly one hospital; each hospital may contain many departments.
Department employs Doctor	Department–Doctor	1:N	Department = partial, Doctor = total	Each doctor works in exactly one department; each department may employ many doctors.
Department Heads Doctor	Department–Head Doctor	1:1	Department = total, Doctor = partial	Each department has one head doctor; not every doctor is a department head.
Doctor supervises Doctor	Doctor–Doctor (supervision)	1:N	Both roles = partial	A doctor may supervise other doctors, and some doctors may have no supervisor.
Patient has Appointment	Patient–Appointment	1:N	Patient = partial, Appointment = total	A patient may have multiple appointments; each appointment belongs to exactly one patient.
Doctor attends Appointment	Doctor–Appointment	1:N	Doctor = partial, Appointment = total	A doctor may have many appointments; each appointment belongs to exactly one doctor.
Appointment prescribes Medication	Appointment–Medication	M:N	Both = partial	An appointment may prescribe multiple medications, and a medication may appear in many appointments. Relationship attributes: Dosage, Duration.

Task 3 — Weak Entity Analysis

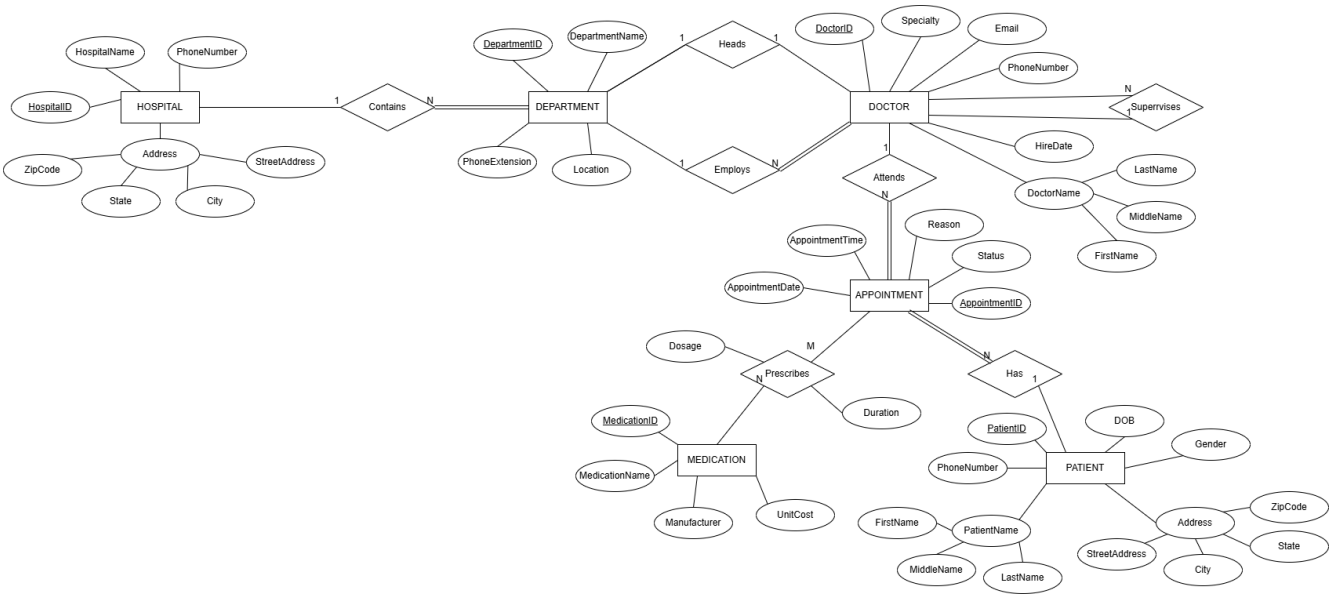
Department

Department should be modeled as a strong entity. Its name is not globally unique because department names are unique only within a hospital. However, assigning DepartmentID gives it an independent identifier. Although every department belongs to one hospital, it does not need an identifying relationship for identification if DepartmentID is used. Therefore, Department is not modeled as a weak entity.

Appointment

Appointment should also be modeled as a strong entity. It is related to one patient and one doctor, but that business dependency does not require weak-entity treatment. AppointmentID gives each appointment a globally unique identifier, which makes it independently identifiable. Therefore, Appointment does not require an identifying relationship and is modeled as a strong entity.

Task 4 — Final ER Diagram



Task 5 — Database Schema and Instance examples

HOSPITAL (<u>HospitalID</u>, HospitalName, StreetAddress, City, State, ZipCode, PhoneNumber)

DEPARTMENT (<u>DepartmentID</u>, DepartmentName, Location, PhoneExtension, HospitalID, HeadDoctorID)

DOCTOR (<u>DoctorID</u>, FirstName, MiddleName, LastName, Specialty, Email, PhoneNumber, HireDate, DepartmentID, SupervisorID)

PATIENT (<u>PatientID</u>, FirstName, MiddleName, LastName, DOB, Gender, StreetAddress, City, State, ZipCode, PhoneNumber)

APPOINTMENT (<u>AppointmentID</u>, AppointmentDate, AppointmentTime, Reason, Status, PatientID, DoctorID)

MEDICATION (<u>MedicationID</u>, MedicationName, Manufacturer, UnitCost)

PRESCRIPTION (<u>AppointmentID</u>, MedicationID, Dosage, Duration)

HOSPITAL

HospitalID	HospitalName	City	State	PhoneNumber
H1	Valley Regional Hospital	Seattle	WA	206-555-1000
H2	Northside Medical Center	Tacoma	WA	253-555-2000

H3	Riverbend Hospital	Spokane	WA	509-555-3000

DEPARTMENT

DepartmentID	DepartmentName	Location	HospitalID	HeadDoctorID
D1	Cardiology	Floor 2	H1	DR1
D2	Pediatrics	Floor 3	H1	DR2
D3	Oncology	Wing B	H2	DR3
D4	Neurology	Floor 4	H3	DR4

DOCTOR

DoctorID	FirstName	LastName	Specialty	DepartmentID	SupervisorID
DR1	Emily	Carter	Cardiology	D1	NULL
DR2	James	Patel	Pediatrics	D2	DR1
DR3	Sarah	Kim	Oncology	D3	NULL
DR4	Daniel	Lopez	Neurology	D4	DR3

PATIENT

PatientID	FirstName	LastName	DOB	Gender	City
P1	Olivia	Reed	1990-04-12	F	Seattle
P2	Noah	Green	1985-07-20	M	Tacoma
P3	Ava	Brown	2001-11-05	F	Spokane
P4	Liam	White	1978-02-14	M	Seattle

APPOINTMENT

AppointmentID	AppointmentDate	AppointmentTime	Reason	PatientID	DoctorID
A1	2026-05-10	09:00 AM	Chest pain	P1	DR1
A2	2026-05-12	11:30 AM	Child checkup	P2	DR2
A3	2026-05-13	02:15 PM	Follow-up	P3	DR3
A4	2026-05-14	10:00 AM	Headache evaluation	P4	DR4

MEDICATION

MedicationID	MedicationName	Manufacturer	UnitCost
M1	Aspirin	Bayer	5.50
M2	Amoxicillin	Pfizer	12.00
M3	Ibuprofen	Advil	7.25
M4	Metformin	Merck	15.75

PRESCRIPTION

AppointmentID	MedicationID	Dosage	Duration
A1	M1	100 mg	7 days
A2	M2	250 mg	10 days

A3	M4	500 mg	30 days
A4	M3	200 mg	5 days